Eigenvalue and Eigenvector

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Determinant 行列式

- Given a **square matrix** <u>A</u> its **determinant** is a real number associated with the matrix.
- The determinant of <u>A</u> is written:

$$\det(\underline{A}) \qquad \text{or} \qquad |\underline{A}|$$

- For a 2x2 matrix, the definition is

  - 若 $\det(A) > 0$：线性变换保留方向；
  - 若 $\det(A) < 0$：变换翻转了方向；
  - 若 $\det(A) = 0$：行（或列）线性相关，矩阵不可逆（无逆矩阵）。

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

- For larger matrices the definition is more complicated

Determinant of a 3x3 Matrix

Finding the Determinant of a Three-By-Three Matrix

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$\det(A) = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$$

Determinants 2x2 Examples

- Example: Calculate det

$$\det\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1\text{x}4 - 2\text{x}3 = -2$$

$$\det\begin{pmatrix} -5 & 2 \\ -2 & 0 \end{pmatrix} = \begin{vmatrix} -5 & 2 \\ -2 & 0 \end{vmatrix} = (-5)\text{x}0 - 2\text{x}(-2) = 4$$

Eigenvalue and Eigenvector特征值与特征向量

特征值问题

- Eigenvalue problem (one of the most important problems in the linear algebra): 线性代数

  If A is an $n \times n$ matrix, do there exist nonzero vectors $\mathbf{x}$ in R^n such that $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$?

  (The term eigenvalue is from the German word *Eigenwert*, meaning "proper value")

- Eigenvalue and Eigenvector:

  A: an $n \times n$ matrix

  λ: a scalar (could be **zero**)

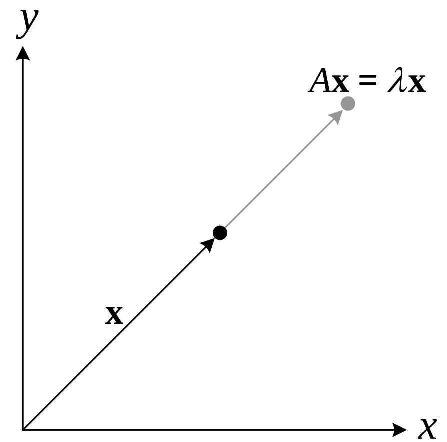
  $\mathbf{x}$: a **nonzero** vector in R^n

Eigenvalue

$$A\mathbf{x} = \lambda\mathbf{x}$$

Eigenvector

※ Geometric Interpretation



Eigenvalue and Eigenvector

- Example: Verifying eigenvalues and eigenvectors

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Eigenvalue

$$A\mathbf{x}_1 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2\mathbf{x}_1$$

Eigenvector

Eigenvalue and Eigenvector

- Exercise: Verifying eigenvalues and eigenvectors

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Eigenvalue and Eigenvector

- Exercise: Verifying eigenvalues and eigenvectors

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Eigenvalue

$$A\mathbf{x}_2 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (-1)\mathbf{x}_2$$

Eigenvector

Characteristic Equation 特征方程

- **Theorem: Finding eigenvalues and eigenvectors of a matrix $A \in M_{n\times n}$**

  Let A be an $n\times n$ matrix.

  (1) An eigenvalue of A is a scalar λ such that det(A-λI)=0

  (2) The eigenvectors of A corresponding to λ are the nonzero solutions of $(A-\lambda I)\mathbf{x}=0$

- Note: follwing the definition of the eigenvalue problem

  $(A-\lambda I)\mathbf{x}=0$ has nonzero solutions for $\mathbf{x}$ iff det(A-λI)=0

if and only if (shortened iff) is a biconditional logical connective between statements

- Characteristic equation of A:

  det(A-λI)=0

$A = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$

$A - \lambda I = \begin{pmatrix} 2-\lambda & 0 \\ 0 & -1-\lambda \end{pmatrix}$

行列式：

$\det(A-\lambda I) = (2-\lambda)(-1-\lambda)$

令其等于 0：

$(2-\lambda)(-1-\lambda) = 0 \Rightarrow \lambda = 2,\ -1$

✅ 所以特征值为 $\lambda_1 = 2$, $\lambda_2 = -1$。

再代入各自的 $(A-I)x=0$，即可求得对应特征向量。

Orthonormal Matrix 正交归一矩阵

- Orthonormal matrix:

  A square matrix P is called orthonormal if it is invertible and

$$P^{-1} = P^{T} \text{ (or } PP^{T} = P^{T}P = I\text{)}$$

- Theorem: Properties of orthonormal matrices

  An $n \times n$ matrix P is orthogonal if and only if its column vectors form an orthonormal set 一个 n * n 矩阵 P 是正交的，当且仅当它的列向量构成一个**标准正交集**（orthonormal set）。

Pf: Suppose the column vectors of P form an orthonormal set,

$$PP^{T} = \begin{bmatrix} p_{11} & \Lambda & & p_{1m} \\ & \Lambda & & \\ \Lambda & \Lambda & \Lambda & \Lambda \\ p_{m1} & \Lambda & & p_{mm} \end{bmatrix} = I$$

Orthonormal Matrix

- Example: Show that P is an orthonormal matrix.

$$P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-2}{3\sqrt{5}} & \frac{-4}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix}$$

Solution: If P is a orthonormal matrix, $P^{-1} = P^T \Rightarrow PP^T = I$

then

$$PP^T = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-2}{3\sqrt{5}} & \frac{-4}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{-2}{\sqrt{5}} & \frac{-2}{3\sqrt{5}} \\ \frac{2}{3} & \frac{1}{\sqrt{5}} & \frac{-4}{3\sqrt{5}} \\ \frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Orthonormal Matrix

Moreover, let $\mathbf{p}_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{-2}{\sqrt{5}} \\ \frac{-2}{3\sqrt{5}} \end{bmatrix}$, $\mathbf{p}_2 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{\sqrt{5}} \\ \frac{-4}{3\sqrt{5}} \end{bmatrix}$, and $\mathbf{p}_3 = \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{5}{3\sqrt{5}} \end{bmatrix}$,

we can produce $\mathbf{p}_1 \cdot \mathbf{p}_2 = \mathbf{p}_1 \cdot \mathbf{p}_3 = \mathbf{p}_2 \cdot \mathbf{p}_3 = 0$ and $\mathbf{p}_1 \cdot \mathbf{p}_1 =$ $\mathbf{p}_2 \cdot \mathbf{p}_2 = \mathbf{p}_3 \cdot \mathbf{p}_3 = 1$